Slide Set #6

# CLRM: further topics

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Notes

# PART I

CONNECTIONS BETWEEN OLS AND MLE



Notes

## Conditional and unconditional modeling

Suppose we want to estimate the following additive regression model

$$\boldsymbol{y} = f(\boldsymbol{X}; \boldsymbol{\beta}) + \boldsymbol{\varepsilon}$$

with the unknown parameter vector $\boldsymbol{\theta} = (\boldsymbol{\beta}, \sigma^2)$.

Two cases:

- Unconditional modeling: we treat $\boldsymbol{X}$ as random as it is the unobservable $\varepsilon$. We do not assume knowledge about *information* generated by $\boldsymbol{X}$.
- Conditional modeling: we treat $\boldsymbol{X}$ as random, but we assume knowledge of the *information generated* by $\boldsymbol{X}$. We still do not assume knowledge of the information generated by $\varepsilon$.

**Remark:** the model based on A1–A5 is a conditional model. We developed conditional inference.



Notes

In 2nd part of the course we consider large sample inference for a more general unconditional linear model. Why large samples?

Consider the OLS for the CLRM, we showed $\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}] = \boldsymbol{\beta}$

Apply the "*law of total variance*" formula[1]

$$\begin{aligned}
\mathrm{Var}[\boldsymbol{b}] &= \mathrm{E}[\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]] + \mathrm{Var}[\mathrm{E}[\boldsymbol{b} \,|\, \boldsymbol{X}]] \\
&= \mathrm{E}[\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]] + \mathrm{Var}[\boldsymbol{\beta}] \\
&= \mathrm{E}[\sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}] + 0 \\
&= \sigma^2 \, \mathrm{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}]
\end{aligned}$$

### Unconditional inference with OLS

In order to use OLS to make *unconditional inference*, we need to know $\mathrm{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}]$. We will use large sample theory to estimate these population moments using WLLN-type theorems

[1] see: https://en.wikipedia.org/wiki/Law_of_total_variance



Notes

## Maximum Likelihood (ML)

Likelihood inference requires knowledge of the *joint density* of the *observable random variables* of the model.

Let $h(\boldsymbol{y}; \boldsymbol{X}; \boldsymbol{\theta})$ be the joint density of $\boldsymbol{y}$,and $\boldsymbol{X}$s, then

$$\hat{\boldsymbol{\theta}}_{\text{ML}} := \arg\max_{\boldsymbol{\theta}} \ \log(h(\boldsymbol{y}; \boldsymbol{X}; \boldsymbol{\theta}))$$

Usually it is assumed that

- $\boldsymbol{y} \,|\, \boldsymbol{X}$ has joint density $f(\boldsymbol{y} \,|\, \boldsymbol{X}; \boldsymbol{\theta})$
- $\boldsymbol{X}$ has joint density $g(\boldsymbol{X})$ *not depending* on the unknown $\boldsymbol{\theta}$

This allows to factorize the joint density:

$$h(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta}) = f(\boldsymbol{y} \,|\, \boldsymbol{X}; \boldsymbol{\theta}) g(\boldsymbol{X})$$



Notes

Which implies

$$\log(h(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta})) = \log(f(\boldsymbol{y} \,|\, \boldsymbol{X}; \boldsymbol{\theta})) + \log(g(\boldsymbol{X}))$$

Therefore

$$\arg\max_{\boldsymbol{\theta}} \ \log(h(\boldsymbol{y}; \boldsymbol{X}; \boldsymbol{\theta})) = \arg\max_{\boldsymbol{\theta}} \ \log(f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\theta}))$$

Thus, ML under the unconditional (joint) model is equivalent to the one obtained under the conditional specification.

Now we will show that the connection between the OLS and ML for the CLRM model under (A1)–(A5).



Notes

## ML for the CLRM + Gaussian errors

Under (A5)

$$\boldsymbol{\varepsilon}|\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n) \implies \boldsymbol{y}\,|\,\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_n)$$

here $f(\boldsymbol{y}, \boldsymbol{X}; \boldsymbol{\beta})$ is the multivariate normal density function with mean $\boldsymbol{X}\boldsymbol{\beta}$ and varcov matrix $\sigma^2 \boldsymbol{I}_n$

$$f(\boldsymbol{y}\,|\,\boldsymbol{X}; \boldsymbol{\theta}) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})\right)$$

and

$$\begin{aligned}\log(f(\boldsymbol{y}\,|\,\boldsymbol{X}; \boldsymbol{\theta})) &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \\ &\propto -n\log(\sigma^2) - \frac{1}{\sigma^2} S(\boldsymbol{\beta})\end{aligned}$$



Notes

therefore

$$\hat{\boldsymbol{\theta}}_{\mathsf{ML}} := \arg\max_{\boldsymbol{\theta}} \left(-n\log(\sigma^2) - \frac{1}{\sigma^2} S(\boldsymbol{\beta})\right)$$

The optimization problem is separable wrt $\boldsymbol{\beta}$ and $\sigma^2$, hence

$$\arg\max_{\boldsymbol{\beta}} \left(-n\log(\sigma^2) - \frac{1}{\sigma^2} S(\boldsymbol{\beta})\right) = \arg\min_{\boldsymbol{\beta}} S(\boldsymbol{\beta})$$

$\implies \hat{\boldsymbol{\beta}}_{\mathsf{ML}} = \hat{\boldsymbol{\beta}}_{\mathsf{OLS}}$

It can be shown that

$$\hat{\sigma}^2_{\mathsf{ML}} = \frac{1}{n}\boldsymbol{e}'\boldsymbol{e} = \frac{1}{n}\sum_{i=1}^{n} e_i^2$$



Notes

Under CLRM + A5 (normal and spherical errors)

- for coefficients ML≡OLS,
- ML for $\sigma^2$ is biased (in finite samples)
- Cramer-Rao inequality: under regularity conditions, the ML attains the variance lower bound



Notes

# PART II

TYPICAL VIOLATIONS THE OF CLRM



Notes

# Multicollinearity

Notes

### Perfect multicollinearity

a regressor is an exact linear copy of another regressor, therefore the design matrix $\boldsymbol{X}$ is rank-deficient, and $\boldsymbol{X}'\boldsymbol{X}$ cannot be inverted

### Multicollinearity

there regressors highly correlated each other. The effects are:

- the scatter of the regressors is highly concentrated, which means that $\det(\boldsymbol{S}_{\boldsymbol{xx}})$ is close to zero, and computation of $(\boldsymbol{X}'\boldsymbol{X})^{-1}$ may be numerically unstable
- whatever depends on $(\boldsymbol{X}'\boldsymbol{X})^{-1}$ (e.g. standard errors, default tests, etc) is not reliable

Multicollinearity is essentially a numerical problem, but in practice there is a clear intuition about it



Notes

The effect is worse if the highly correlated regressors have also both a dominant marginal effect on the $Y$:

$$\texttt{Earning} = \beta_1 + \beta_2 \texttt{Age} + \beta_3 \texttt{Experience} + \beta_4 \texttt{Scholing} + \ldots$$

it may be that $\mathrm{Cor}(\texttt{Age}, \texttt{Experience}) =$ large, this means that it is difficult to identify their role precisely $\implies$ *low precision = large standard errors*

### Detection (some rules of thumb)

- compute solve(det(t(X)%*%X)), if it is too low (below $2 \times 10^{16}$) you may run into troubles
- compute kappa(t(X)%*%X) (= condition number of $\boldsymbol{X}'\boldsymbol{X}$), if it is larger than 50 you may run into troubles

Unfortunately absolute *safe lower-bounds* don't exist!

### Cures

regularization and variable transformations (not treated here)

## Heteroskedasticity

In (A4) we assume $\mathrm{Var}[\boldsymbol{\varepsilon} \mid \boldsymbol{X}] = \sigma^2 \boldsymbol{I}_n$:

- the diagonal of $\mathrm{Var}[\boldsymbol{\varepsilon} \mid \boldsymbol{X}]$ is constant
- $\mathrm{Var}[\varepsilon_i \mid \boldsymbol{X}] = \sigma^2$ for all $i$ (homoskedasticity)

Under (A4) the dispersion around the regression function

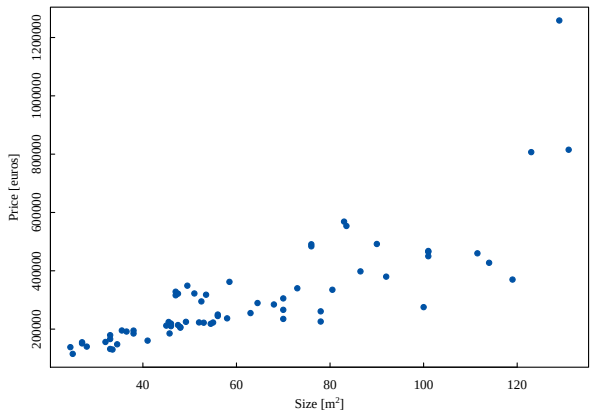
- cannot depend on $i$, the unit index
- cannot depend on the level of one or more regressors

Homoskedasticity is a strong restriction often violated in econometric models



Notes

**Example:** "*Espoo housing price data*"



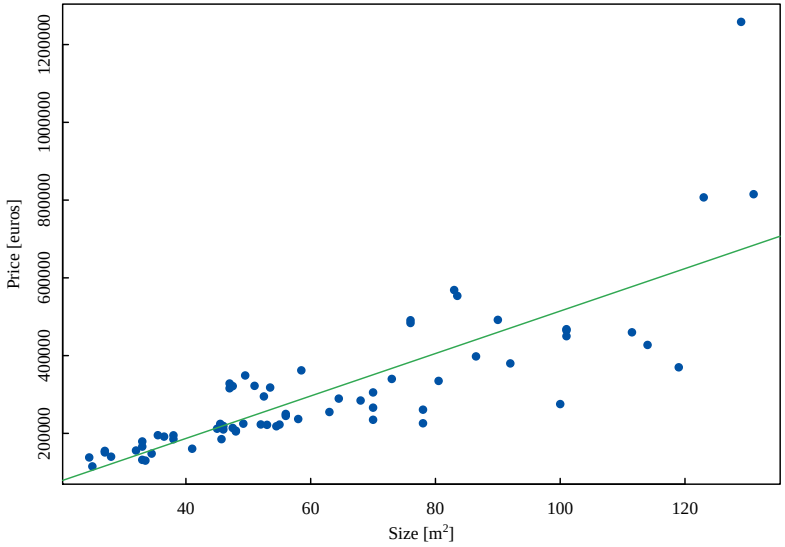

We perform OLS to estimate the following model

$$\texttt{Price}_i = \beta_1 + \beta_2 \texttt{Size}_i + \varepsilon_i$$



Notes

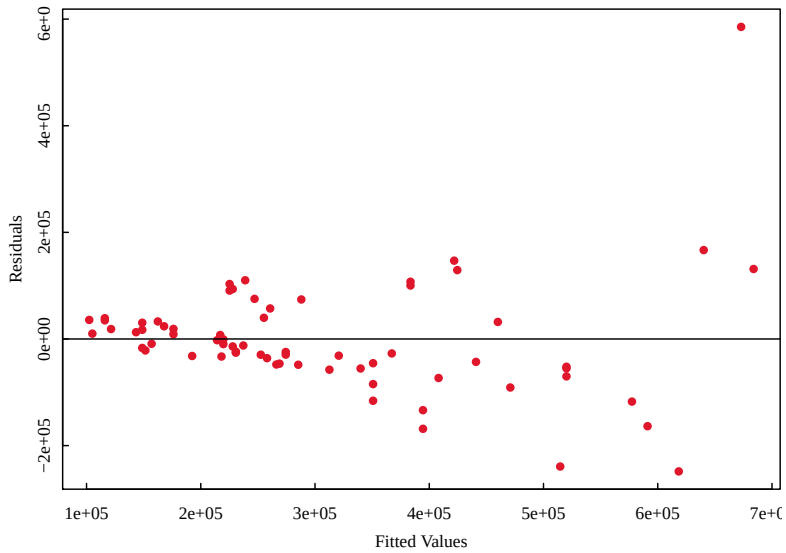
## Estimated Regression Function






Notes

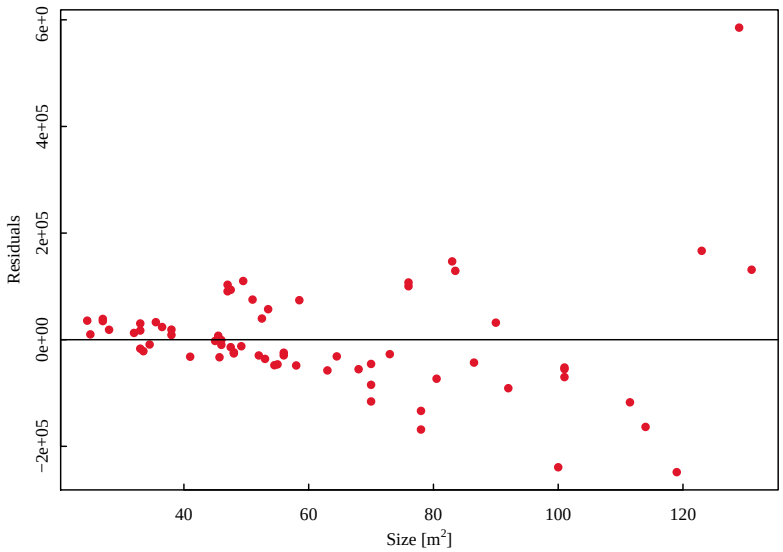
## Visual Diagnostic: Residuals-vs-Fitted plot






Notes

The source of the heteroskedasticity is because the variance of the error depends on the level of `Size`





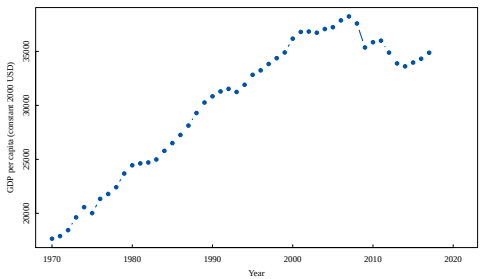

Notes

## Lagged-variables and serial correlation

**Lag**

$X_t$ is a RV sampled at time $t$, and $X_{t-l}$ its measurement $l$ periods before. $l$ is called *lag*.

Suppose $X =$ `Italy's per capita GDP` measured in 2000-USD from 1970 to 2018

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Notes

Imagine to construct a data matrix with three regressors $X_{t-2}, X_{t-1}, X_t$

$$\boldsymbol{X} = \begin{pmatrix} \texttt{GDP}_{1970} & \texttt{GDP}_{1971} & \texttt{GDP}_{1972} \\ \texttt{GDP}_{1971} & \texttt{GDP}_{1972} & \texttt{GDP}_{1973} \\ \vdots & \vdots & \vdots \\ \texttt{GDP}_{2017} & \texttt{GDP}_{2018} & \texttt{GDP}_{2019} \end{pmatrix}$$

Serial correlation (aka autocorrelation) at lag $l$

$$\rho(l) = \mathrm{Cor}[X_t, X_{t-l}]$$

For the GDP example, $\rho(2)$ is the correlation between the first and the third column of $\boldsymbol{X}$

Non-zero serial correlations may be be produced by *autoregressive linear models*

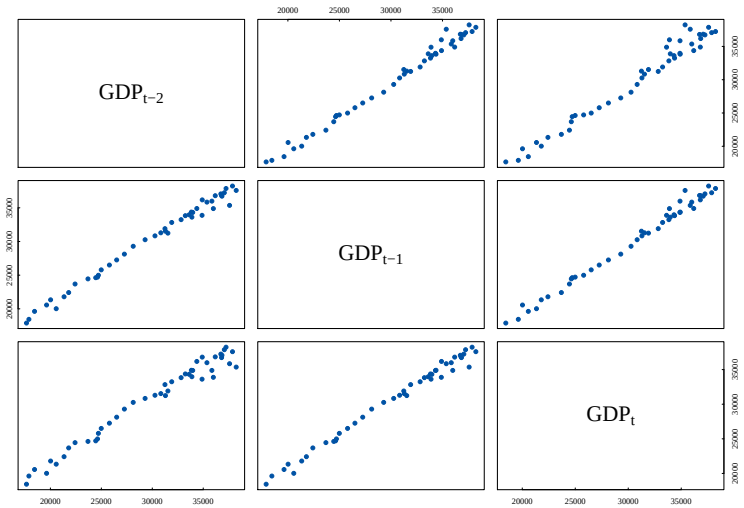
$$X_t = \theta_0 + \theta_1 X_{t-1} + \theta_1 X_{t-2} + \ldots + \text{error term}$$

i.e. regressors are lagged versions of the dependent variable



Notes

pairplot from the GDP data matrix $\boldsymbol{X}$






Notes

## Autocorrelated errors

In A4 we assume $\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] = \sigma^2 \boldsymbol{I}_n$:

- off-diagonal elements of $\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}]$ are all zero
- $\mathrm{Cov}[\varepsilon_i, \varepsilon_j] = 0$ for all $i \neq j$

Example

- $i = t = \texttt{time}$, regressors are measured as time series
- there are one or more lagged regressors

Consider the *demand model*

$$\texttt{Quantity}_t = \beta_1 + \beta_2 \texttt{Price}_t + \beta_2 \texttt{Income}_{t-1} + \ldots + \varepsilon_t \tag{6.1}$$

Under (A4) $\mathrm{Cor}[\varepsilon_t, \varepsilon_{t-l}] = 0$ for all $t$ and $l$.

(A4) implies absence of serial correlation in the error. This is often violated when we have time series sampling



Notes

Example: "*Demand for Ice Cream*" (Hildreth and Lu, 1960)
weekly observations from 18 March 1951 to 11 July 1953 on

$C_t$ consumption of ice cream per head (in pints)
$I_t$ average family income per week (in USD)
$P_t$ price of ice cream (USD/pint)
$X_t$ average temperature (in Fahrenheit)

Consider the "*demand of ice cream*" model

$$C_t = \beta_0 + \beta_1 I_t + \beta_2 P_t + \beta_3 X_t + \varepsilon_t \tag{6.2}$$

The regression function gives the expected aggregate "expenditure" for ice cream given the current information. $\varepsilon_t$ formalizes the deviation from the expectation

$$\varepsilon_t = C_t - \mathrm{E}[C_t \,|\, \text{Information at time } t]$$



Notes

However, Hildreth and Lu (1960) found that the estimated residuals of (6.2) were correlated at lag 1. What's going on?

The aggregate expenditure measured by $C_t$ reflects a "*rational*" learning process. The decision about $C_t$ is likely to depend on the "*mistake*" $\varepsilon_{t-1}$ that I did in the previous period

Suppose that this happens so that $\varepsilon_t$ correlates with $\varepsilon_{t-1}$:

$$\varepsilon_t = \phi_0 + \phi_1 \varepsilon_{t-1} + u_t \tag{6.3}$$

It's easy to see that *serial correlation* in the error term implies

- $\mathrm{Cov}[\varepsilon_t, \varepsilon_{t-1}] \neq 0$ for all $t > 1$
- $\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}]$ is not diagonal anymore
- $\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}]$ computed based on (A4) is probably wrong!



Notes

## OLS with non-spherical errors

**Assumption (A4-NSE): non-spherical error variance**

Let $\boldsymbol{\Omega} \in \mathbb{R}^{n \times n}$ be a PSD and symmetric matrix, and

$$\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \,|\, \boldsymbol{X}] = \boldsymbol{\Omega}$$

A2 + A4-NSE $\implies$ $\mathrm{Var}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] = \boldsymbol{\Omega}$ is non-spherical.

Under A1, A2, A3, and A4-NSE, the OLS estimator $\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$ is still unbiased (homework), but

$$\mathrm{Var}[\boldsymbol{b} \,|\, \boldsymbol{X}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\Omega}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \tag{6.4}$$

Hence, if A4-NSE holds, all the inference methods (standard errors, tests' decision rules, etc) based on previous old formula are wrong



Notes

## Dealing with non-spherical errors

In the old days there were a number of techniques to test, or correct for specific forms of heteroskedasticity or serial correlation. Today there are couple of options

### Finite sample methods

- Assume that $\boldsymbol{\Omega} = g(\boldsymbol{X})$ is known, use the Generalized Least Squares Estimator (GLS) (homework)
- Under A4-NSE + Gaussian errors, the inference based on GLS will be very similar to what we have seen for OLS

*Problem:* $\boldsymbol{\Omega}$ is not known, and normality often fails!

### Large sample methods

- consider a setup that accounts for non-spherical errors
- Gaussianity is not needed because of large sample approximations

*Problem:* sometimes $n$ is not sufficiently large



Notes

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